

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

- Improved throughput and reliability for high-frequency defense sensor data pipelines by engineering distributed Python and C++ microservices on AWS, designing storage schemas and retrieval patterns optimized for high-volume, low-latency production workloads.
- Reduced data inconsistency across distributed ingestion services by applying rigorous data modeling and fault-isolation patterns, ensuring integrity and queryability of sensor data across multiple heterogeneous hardware sources.
- Increased deployment confidence across production targets by building automated CI/CD infrastructure with static analysis, integration testing, and infrastructure-as-code tooling that enforced consistency at scale.

Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

- Improved clinical data completeness and downstream analytics accuracy by designing and operating Python and TypeScript data pipelines on AWS SageMaker and Fargate that ingested unstructured WebRTC session data, ran LLM inference, and persisted structured outputs to PostgreSQL with full lineage and auditability.
- Reduced manual data preparation overhead by building automated ingestion and transformation workflows that normalized heterogeneous clinical inputs into a consistent, queryable data model consumed by provider-facing dashboards and AI downstream systems.
- Enabled zero-downtime releases and reproducible data environments by owning YAML-based CI/CD configuration and containerized AWS infrastructure, maintaining data integrity across production deployments.

Projects

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.